

"Challenge_NotchItUp"

Analyst tool: Volatility 3 Framework 2.28.0

Memory image: Challenge.raw (1,610,547,200 bytes, ~1.6 GB)

Host profile: Windows 7 SP1 x64 (Win7SP1x64), single CPU

Image acquisition timestamp: 2019-08-19 14:41:58 UTC (SystemTime per windows.info)

Prepared By: Dnyaneshwari Kale

1. Objective

Analyze the provided memory dump to recover two hidden CTF flags related to a password-protected RAR archive found on the suspect's desktop.

2. Environment & Tool Setup

Before running Volatility plugins, the working directory containing the memory image and Volatility binaries was inspected to confirm the tools and target file were present.

```
PS C:\...\MemoryForensics\Backup> cd ..
PS C:\...\MemoryForensics\Backup> dir

Challenge_NotchItUp\    (folder)
symbols\              (folder)
Challenge_NotchItUp.7z
vol.exe
volatility_2.6_win64_standalone.exe
volshell.exe

PS ...\Backup> .\volatility_2.6_win64_standalone.exe -f
.\Challenge_NotchItUp\Challenge.raw
--profile=Win7SP1x64 pslist
```

```
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> cd ..
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup> dir

Directory: C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup

Mode                LastWriteTime         Length Name
----                -
d-----            8/23/2026 10:36 PM             Challenge_NotchItUp
d-----            8/22/2026 11:19 PM             symbols
-a-----            5/31/2020 1:45 PM              0 .ventoyignore
-a-----            8/21/2026 12:07 PM      359074541 Challenge_NotchItUp.7z
-a-----            7/22/2026 12:02 PM      22216348 vol.exe
-a-----            7/31/2026 12:28 AM      15794079 volatility_2.6_win64_standalone.exe
-a-----            7/22/2026 12:02 PM      22292045 volshell.exe

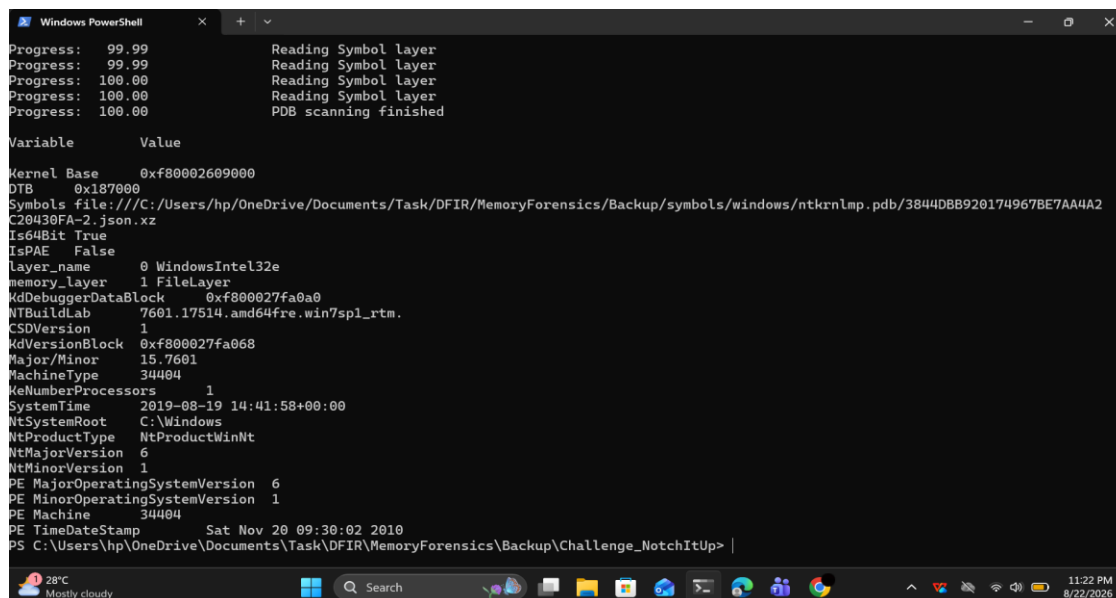
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup> .\volatility_2.6_win64_standalone.exe -f
.\Challenge_NotchItUp\Challenge.raw --profile=Win7SP1x64 pslist
```

Fig. 1 — Working directory contents and initial Volatility 2.6 pslist invocation.

3. Confirming the Memory Image Profile

windows.info was run against the raw image to confirm the correct OS build and symbol table before further analysis.

```
..\vol.exe -f .\Challenge.raw windows.info
```



```
Progress: 99.99      Reading Symbol layer
Progress: 99.99      Reading Symbol layer
Progress: 100.00     Reading Symbol layer
Progress: 100.00     Reading Symbol layer
Progress: 100.00     PDB scanning finished

Variable           Value
-----
Kernel Base        0xf80002609000
DTB                0x187000
Symbols file:////C:/Users/hp/OneDrive/Documents/Task/DFIR/MemoryForensics/Backup/symbols/windows/ntkrnlmp.pdb/38440BB920174967BE7AA4A2C20430FA-2.json.xz
Is64Bit            True
IsPAE              False
Layer_name         0 WindowsIntel32e
memory_layer       1 FileLayer
KdDebuggerDataBlock 0xf800027fa0a0
NTBuildLab         7601.17514.amd64fre.win7sp1_rtm.
CSDVersion         1
KdVersionBlock     0xf800027fa068
Major/Minor        15,7601
MachineType        34404
KeNumberProcessors 1
SystemTime         2019-08-19 14:41:58+00:00
NtSystemRoot       C:\Windows
NtProductType       NtProductWinNt
NtMajorVersion     6
NtMinorVersion     1
PE MajorOperatingSystemVersion 6
PE MinorOperatingSystemVersion 1
PE Machine         34404
PE TimeDateStamp   Sat Nov 20 09:30:02 2010
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\Memo...>
```

Fig. 2 — windows.info output: Kernel Base 0xf80002609000, DTB 0x187000, NTBuildLab 7601.17514.amd64fre.win7sp1_rtm — confirming Win7 SP1 x64.

4. Process Enumeration

windows.pslist, windows.psscan, and windows.pstree were run to build a picture of running/terminated processes and parent-child relationships.

```
..\vol.exe -f .\Challenge.raw windows.pslist
```

```
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.pslist
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
4 0 System 0xfa80012a5040 78 495 N/A False 2019-08-19 14:40:07.000000 UTC N/A Disabled
264 4 smss.exe 0xfa8002971470 2 29 N/A False 2019-08-19 14:40:07.000000 UTC N/A Disabled
336 328 csrss.exe 0xfa800234cb30 10 415 0 False 2019-08-19 14:40:10.000000 UTC N/A Disabled
384 328 wininit.exe 0xfa8002aae910 3 74 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
396 376 csrss.exe 0xfa8002ab7060 9 499 1 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
436 376 winlogon.exe 0xfa8002b66560 6 116 1 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
480 384 services.exe 0xfa8002b99200 9 194 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
496 384 lsass.exe 0xfa8002bb4600 7 513 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
504 384 lsm.exe 0xfa80022ff910 10 152 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
608 480 svchost.exe 0xfa8002ce8740 10 358 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
668 480 VBoxService.ex 0xfa8002d13060 13 136 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
724 480 svchost.exe 0xfa8002d4bb30 6 257 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
780 480 svchost.exe 0xfa8002d4fb30 19 405 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
896 480 svchost.exe 0xfa8002dcf5f0 22 452 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
948 480 svchost.exe 0xfa8002delb30 35 893 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
1008 780 audiodg.exe 0xfa8002e0b1c0 7 132 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
400 480 svchost.exe 0xfa8002e645f0 13 275 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
1052 480 svchost.exe 0xfa8002eac740 17 368 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
1176 480 spoolsv.exe 0xfa8002e76b30 14 279 0 False 2019-08-19 14:40:13.000000 UTC N/A Disabled
1212 480 svchost.exe 0xfa8002f4d780 21 311 0 False 2019-08-19 14:40:13.000000 UTC N/A Disabled
1308 480 svchost.exe 0xfa8002f79b30 17 253 0 False 2019-08-19 14:40:13.000000 UTC N/A Disabled
1812 480 taskhost.exe 0xfa8003144250 9 147 1 False 2019-08-19 14:40:18.000000 UTC N/A Disabled
1868 896 dwm.exe 0xfa8003160120 4 70 1 False 2019-08-19 14:40:18.000000 UTC N/A Disabled
1876 948 taskeng.exe 0xfa8003164b30 5 81 0 False 2019-08-19 14:40:18.000000 UTC N/A Disabled
1944 1844 explorer.exe 0xfa800319a060 35 894 1 False 2019-08-19 14:40:19.000000 UTC N/A Disabled
```

Fig. 3 — windows.pslist output listing active processes (PID, PPID, image name, thread/handle counts).

```
..\vol.exe -f .\Challenge.raw windows.psscan
```

```
Windows PowerShell
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.psscan
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
2080 3060 firefox.exe 0x5d905b30 59 970 1 True 2019-08-19 14:41:08.000000 UTC N/A Disabled
2440 2124 chrome.exe 0x5db41b30 13 263 1 False 2019-08-19 14:40:54.000000 UTC N/A Disabled
2452 2124 chrome.exe 0x5db4bb30 14 167 1 False 2019-08-19 14:40:54.000000 UTC N/A Disabled
864 2256 GoogleCrashHan 0x5dde3700 1 1279459345 0 False 2019-08-19 14:42:41.000000 UTC N/A Disab
led
3716 1944 WinRAR.exe 0x5dde71e0 7 201 1 False 2019-08-19 14:41:43.000000 UTC N/A Disabled
924 1928 GoogleCrashHan 0x5de19060 6 93 0 False 2019-08-19 14:40:19.000000 UTC N/A Disabled
1292 1928 GoogleCrashHan 0x5de27060 7 105 0 True 2019-08-19 14:40:19.000000 UTC N/A Disabled
1108 1944 VBoxTray.exe 0x5de77810 14 139 1 False 2019-08-19 14:40:20.000000 UTC N/A Disabled
2940 2124 chrome.exe 0x5ded4380 9 172 1 False 2019-08-19 14:41:06.000000 UTC N/A Disabled
2292 608 WmiPrivSE.exe 0x5ded9060 13 288 0 False 2019-08-19 14:40:52.000000 UTC N/A Disabled
2340 2124 chrome.exe 0x5def9a70 12 282 1 False 2019-08-19 14:40:52.000000 UTC N/A Disabled
856 480 SearchIndexer. 0x5df15060 13 689 0 False 2019-08-19 14:40:27.000000 UTC N/A Disabled
1812 480 taskhost.exe 0x5e144250 9 147 1 False 2019-08-19 14:40:18.000000 UTC N/A Disabled
2168 2124 chrome.exe 0x5e14fab0 3 55 1 False 2019-08-19 14:40:49.000000 UTC N/A Disabled
1868 896 dwm.exe 0x5e160120 4 70 1 False 2019-08-19 14:40:18.000000 UTC N/A Disabled
1876 948 taskeng.exe 0x5e164b30 5 81 0 False 2019-08-19 14:40:18.000000 UTC N/A Disabled
1944 1844 explorer.exe 0x5e19a060 35 894 1 False 2019-08-19 14:40:19.000000 UTC N/A Disabled
1008 780 audiodg.exe 0x5e20b1c0 7 132 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
400 480 svchost.exe 0x5e2645f0 13 275 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
1176 480 spoolsv.exe 0x5e276b30 14 279 0 False 2019-08-19 14:40:13.000000 UTC N/A Disabled
1052 480 svchost.exe 0x5e2ac740 17 368 0 False 2019-08-19 14:40:12.000000 UTC N/A Disabled
1212 480 svchost.exe 0x5e34d780 21 311 0 False 2019-08-19 14:40:13.000000 UTC N/A Disabled
1308 480 svchost.exe 0x5e379b30 17 253 0 False 2019-08-19 14:40:13.000000 UTC N/A Disabled
608 480 svchost.exe 0x5e4e8740 10 358 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
668 480 VBoxService.ex 0x5e513060 13 136 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
724 480 svchost.exe 0x5e54bb30 6 257 0 False 2019-08-19 14:40:11.000000 UTC N/A Disabled
```

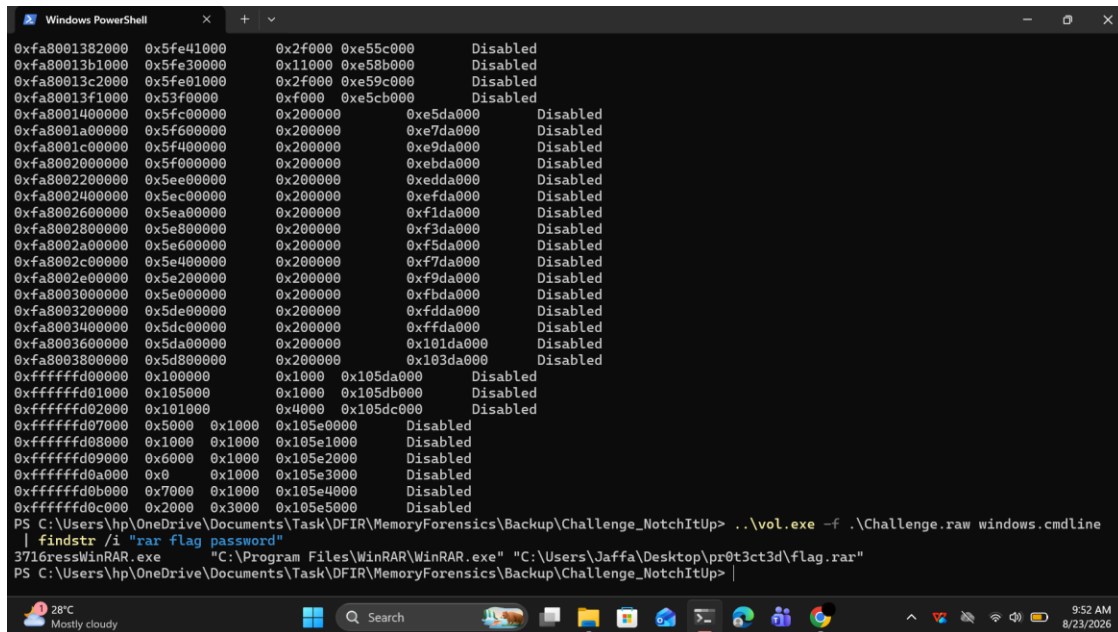
Fig. 4 — windows.psscan output. Notable entry: WinRAR.exe, PID 3716, PPID 1944, created 2019-08-19 14:41:43 UTC.

```
..\vol.exe -f .\Challenge.raw windows.pstree
```



```
..\vol.exe -f .\Challenge.raw windows.cmdline | findstr /i "rar flag password"
```

```
3716 WinRAR.exe "C:\Program Files\WinRAR\WinRAR.exe"  
"C:\Users\Jaffa\Desktop\pr0t3ct3d\flag.rar"
```



```
Windows PowerShell
0xfa8001382000 0x5fe41000 0x2f000 0xe55c000 Disabled
0xfa80013b1000 0x5fe30000 0x11000 0xe58b000 Disabled
0xfa80013c2000 0x5fe01000 0x2f000 0xe59c000 Disabled
0xfa80013f1000 0x53f0000 0xf000 0xe5cb000 Disabled
0xfa8001400000 0x5fc00000 0x200000 0xe5da000 Disabled
0xfa8001a00000 0x5f600000 0x200000 0xe7da000 Disabled
0xfa8001c00000 0x5f400000 0x200000 0xe9da000 Disabled
0xfa8002000000 0x5f000000 0x200000 0xebda000 Disabled
0xfa8002200000 0x5ee00000 0x200000 0xedda000 Disabled
0xfa8002400000 0x5ec00000 0x200000 0xefda000 Disabled
0xfa8002600000 0x5ea00000 0x200000 0xf1da000 Disabled
0xfa8002800000 0x5e800000 0x200000 0xf3da000 Disabled
0xfa8002a00000 0x5e600000 0x200000 0xf5da000 Disabled
0xfa8002c00000 0x5e400000 0x200000 0xf7da000 Disabled
0xfa8002e00000 0x5e200000 0x200000 0xf9da000 Disabled
0xfa8003000000 0x5e000000 0x200000 0xfbda000 Disabled
0xfa8003200000 0x5de00000 0x200000 0xfdda000 Disabled
0xfa8003400000 0x5dc00000 0x200000 0xffda000 Disabled
0xfa8003600000 0x5da00000 0x200000 0x101da000 Disabled
0xfa8003800000 0x5d800000 0x200000 0x103da000 Disabled
0xffffffff000000 0x100000 0x1000 0x105da000 Disabled
0xffffffff010000 0x105000 0x1000 0x105db000 Disabled
0xffffffff020000 0x101000 0x4000 0x105dc000 Disabled
0xffffffff070000 0x5000 0x1000 0x105e0000 Disabled
0xffffffff080000 0x1000 0x1000 0x105e1000 Disabled
0xffffffff090000 0x6000 0x1000 0x105e2000 Disabled
0xffffffff0a0000 0x0 0x1000 0x105e3000 Disabled
0xffffffff0b0000 0x7000 0x1000 0x105e4000 Disabled
0xffffffff0c0000 0x2000 0x3000 0x105e5000 Disabled
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.cmdline
| findstr /i "rar flag password"
3716ressWinRAR.exe "C:\Program Files\WinRAR\WinRAR.exe" "C:\Users\Jaffa\Desktop\pr0t3ct3d\flag.rar"
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> |
```

Fig. 7 — Filtered windows.cmdline output confirming WinRAR.exe (PID 3716) had opened "flag.rar" from C:\Users\Jaffa\Desktop\pr0t3ct3d\.

6. Locating the Archive in Memory

windows.filescan was used to search all file objects referenced in memory, first broadly, then filtered to the target file name.

```
..\vol.exe -f .\Challenge.raw windows.filescan
```

```

Windows PowerShell
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.filesca
n
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Offset Name
0x53f2690 \Endpoint
0x53f2800 \chrome.2968.1.28264847
0x53f3770 \Users\Jaffa\AppData\Local\Mozilla\Firefox\Profiles\84ksw0a.default-release\cache2\entries\F3A2A55211EE66D36F43F15EF
F501E9546680661
0x53f4030 \Users\Jaffa\AppData\Local\Mozilla\Firefox\Profiles\84ksw0a.default-release\cache2\entries\3168C61F89D068684E314E066
8C083341C5929A8
0x53f4700 \Users\Jaffa\AppData\Local\Mozilla\Firefox\Profiles\84ksw0a.default-release\cache2\entries\32027373AB514902694BD2F13
ABE08513EAF1DF9
0x53f5180 \chrome.2968.2.15970394
0x53f5900 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\webappsstore.sqlite-wal
0x53f5d00 \chrome.2968.1.28264847
0x53f5f20 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\favicons.sqlite
0x53f65f0 \Endpoint
0x53f6f20 \chrome.2968.0.66569659
0x53f7070 \Windows\SysWOW64\actxprxy.dll
0x53f7760 \Endpoint
0x53f7cb0 \Users\Jaffa\AppData\Local\Mozilla\Firefox\Profiles\84ksw0a.default-release\cache2\entries\3695979DC4CDB6256489DC66C
1152B1985B55252
0x53f7f20 \chrome.3016.3.194647336
0x53f8070 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\storage\permanent\chrome\idb\165711459
5AmcateirvtiSty.sqlite
0x53f89e0 \Directory
0x53f8d10 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\webappsstore.sqlite
0x53f9070 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\storage\default\https+++mail.google.co
m\cache\morgue\75\{f4ec0805-5e72-4cbe-a262-5a99a7abb04b}.tmp
0x53f94a0 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\webappsstore.sqlite-shm
0x53f9680 \Users\Jaffa\AppData\Roaming\Mozilla\Firefox\Profiles\84ksw0a.default-release\favicons.sqlite-wal

```

Fig. 8 — windows.filescan output (excerpt): file objects referenced in memory, including browser cache and profile artifacts.

```
..\vol.exe -f .\Challenge.raw windows.filescan | findstr /i "flag.rar"
```

```

0x5ff6d570 \Windows\Fonts\vgacem.fon
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.filesca
n | findstr /i "flag.rar"
0x5fcfc4b0 100.0\Users\Jaffa\Desktop\pr0t3ct3d\flag.rar
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp>

```

Fig. 9 — Filtered filescan result: flag.rar located at virtual address 0x5fcfc4b0.

7. Extracting the Archive from Memory

With the virtual address identified, windows.dumpfiles was used to carve the flag.rar file object directly out of the memory image.

```
..\vol.exe -f .\Challenge.raw windows.dumpfiles --virtaddr 0x5fcfc4b0
```

```

PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.filesca
n | findstr /i "flag.rar"
0x5fcfc4b0 100.0\Users\Jaffa\Desktop\pr0t3ct3d\flag.rar
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> ..\vol.exe -f .\Challenge.raw windows.dumpfil
es --virtaddr 0x5fcfc4b0
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Cache FileObject FileName Result
PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp> dir

Directory: C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp

Mode                LastWriteTime         Length Name
----                -
-a-----          8/19/2019   8:12 PM        1610547200 Challenge.raw

PS C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\Backup\Challenge_NotchItUp>

```

Fig. 10 — windows.filescan confirming the offset, followed by windows.dumpfiles extracting flag.rar; dir output confirms Challenge.raw is the working image.

8. Recovering the RAR Password

The extracted flag.rar archive was password-protected. Instead of brute-forcing it, the raw memory image was searched directly for password-related strings left behind in process environment blocks.

```
Searched Challenge.raw for the string "easypeasyvirus"
found within the WinRAR.exe process environment block:
```

```
...cRAR password=easypeasyvirus...
```

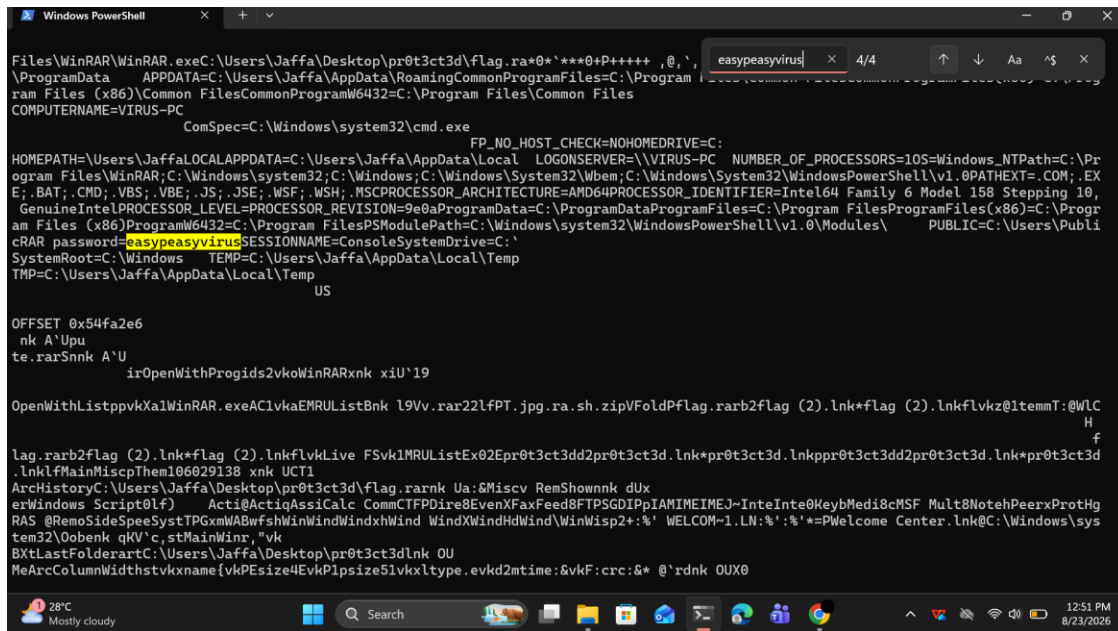
A screenshot of a Windows PowerShell window. The title bar says 'Windows PowerShell'. The window contains a memory dump search result. At the top, a search bar shows 'easypeasyvirus' with '4/4' results. The main text is a memory dump from a WinRAR.exe process. It shows various environment variables like HOMEPath, APPDATA, and COMSPEC. The key finding is 'cRAR password=easypeasyvirus' highlighted in yellow. Below this, there are more memory dump details including offsets and hex values. The taskbar at the bottom shows the date and time as 12:51 PM on 8/23/2026.

Fig. 11 — String search across the memory dump revealing the RAR password "easypeasyvirus" embedded in the process environment strings.

Recovered password: "easypeasyvirus"

```
python -c "from pathlib import Path; b=Path(r'.\Challenge.raw').read_bytes(); import re;
m=re.search(b'Mega.{0,20}Drive.{0,20}Key',b,re.I); print('Offset:',m.start() if m else -1);
print(b[m.start()-2000:m.start()+5000].decode('utf-8','replace') if m else 'Not found')"
```




Fig. 13 — `flag2.png`: "aN_Am4zING!_i_gU3Ss???_}"

10. Final Combined Flag

Concatenating the two recovered fragments in order yields the complete flag:

```
inctf{thi5_cH4LL3Ng3!s_g0nn4_b3_aN_Am4zING!_i_gU3Ss???_}
```

11. Summary of Methodology

Step	Plugin / Action	Purpose
1	windows.info	Identify OS build / profile (Win7 SP1 x64)
2	windows.pslist / psscan / pstree	Enumerate processes; spot WinRAR.exe
3	windows.cmdline	Confirm archive path (flag.rar) opened by WinRAR
4	windows.filescan	Locate the flag.rar file object's virtual address
5	windows.dumpfiles --virtaddr	Extract flag.rar from the raw memory image
6	String search on memory dump	Recover RAR password (easypeasyvirus) from process environment strings
7	Manual archive extraction	Extract flag1.png and flag2.png
8	Visual inspection	Read and combine flag text from both images

12. Conclusion

This exercise demonstrated a full memory forensics workflow: process discovery, artifact identification via command-line arguments, file carving from a live memory image, credential recovery from process environment data, and final evidence extraction — all performed without ever touching a live disk image, using only the volatile memory capture (Challenge.raw).

Final Flag: `inctf{thi5_ch4LL3Ng3_!s_g0nn4_b3_aN_Am4zING_!_i_gU3Ss??_}`